

Exam Solution for TDT4186

Spring 2024

1 Multi-Choice Questions

Automatic grading. Attention: Question 16 is excluded from the total.

1. It consists of a small kernel that provides basic services such as scheduling and inter-process communication, with additional services implemented as user-space processes or servers.
2. A process can only be in one state at a time.
3. Processes are independent and can't communicate with each other. *Processes can communicate with each other using something like pipe.*
4. Processes run independently and are isolated, whereas threads are designed to run as isolated parts of a process.
5. **Round-Robin with time quantum 5** : STCF:23.75 SJF: 23.75 FIFO: 24.5 RR:34.5
6. It aims to provide fair distribution of CPU time among processes using virtual runtime.
7. 2. The if condition checks whether a process is runnable. It can be seen that within the if condition, and there is a context switch
8. A virtual address is generated by the CPU during program execution, while a physical address is assigned by the operating system to map to a specific location in memory.
9. 1,3,4
10. **3,4**: Paging does not improve the processing speed of memory, and multi-level paging reduces the page table size.
11. It caches recently used page table entries to accelerate the translation of virtual addresses to physical addresses.
12. **12, 10**: $4KB=2^{12}$, $1KB=2^{10}$

13. It provides a mechanism for the producer to wait until there is space and for the consumer to wait until there is an item in the buffer.
14. To ensure that only one thread can add or remove items from the buffer at any given time.
15. No. It may freeze because one consumer can signal another consumer even when the buffer is empty.
16. Two semaphores: one initialized to MAX which the producer waits on, and another initialized to 0, which the consumer waits on.
17. All statements are false.
18. Micro-controller
19. To bypass the CPU for certain operations, allowing peripheral devices to access memory directly.
20. HDDs are byte-addressable. **HDDs are sector-addressable.**
21. Cells in an SSD can only be written to a limited number of times.
22. **1 and 4**
23. f=3, if this is the first opened file. **0-2 are reserved for stdio.**
24. 3, 6, 2, 5, 1. **When opening a file, it does not need to touch the data block.**

2 Multi-part questions

For multi-part questions, the automatic grading may fail to award some answers which are slightly different from the provided solutions or written in different phrases. Thus, it is worth noting that all wrong blanks should go through a scrutiny check. Answers that provide a close meaning can be awarded the point without deduction.

Process

Explanation: The first fork() creates two processes, one parent and one child. If the parent executes, the return value is the pid of the child (> 0). It will execute the A part of the ternary operator, i.e., another fork(). Thus, one more process is generated. If the child executes, it will print out success! **Each blank accounts for 1 point. All answers should exactly match the solutions.** The total is **4 points**.

- 2.
- Yes.
- 1. Only the child will print out success.
- Child

Scheduling

Explanation: the first and forth questions may need to be manually evaluated. The total is **7 points**. **If Inspira does not automatically grade this question, the whole question needs to be manually evaluated and added to the total.**

- boost priority. Any phrase expressing the same meaning is acceptable, for example, reset the priority, move tasks to the highest queue, etc.

T1	95	3
T2	60	2
T3	120	3

- No/false. Some may write an explanation in the blank which the automatic grading would take as the wrong result. Hence, pay attention to correct it.

Memory

Explanation: The first question accounts for **3 points**. The remaining two questions each get **1 point**. **All answers need to be manually graded.**

- The answer should consist of two parts: (1) Segmentation divides an address space into several blocks with variable sizes, and flexibly assigns them to physical location. **Two key concepts should be covered in the first part, dividing address space into several blocks/segments, and allocating these segments to different locations in the memory.** If the answer somehow expresses these two concepts, **2 points** can be awarded. (2) Each segmentation has a pair of registers, one for its starting address and one for its size. **The pair of registers should be mentioned to be awarded 1 point.**
- External fragmentation. External fragmentation is the only correct answer. **If both internal and external fragmentation are mentioned, no point is given.**
- Compact. **'Compact'** is the exact terminology that should be used. However, if the answer implies that small memory holes/blocks are combined to generate a large memory block. It can be awarded **1 point**.

Page

Explanation: The total is **8 points**. Each blank account for **1 point**.

- 9 or 27. Each page directory can store $2^{12}/2^3 = 2^9$, for each level it needs 9 bits. If you answer 27, it will be considered as a correct answer.
- The binary of 0X23059C20AF is 0010001100000101100111000010000010101111. Three-level has 9 bits, 9bits, 9bits, and 12bits. It can be divided into

010001100 — — — 000101100 — — — 111000010 — — — 000010101111

The answers that omit unnecessary 0s in the most significant bits can be accepted as the correct ones. Others should be graded as wrong. For example of 010001100, 10001100 is correct, but 01000110 is wrong.

- 512GB: 2^{39} . Either 512 GB or 2^{39} can be graded as correct.
- FIFO: FIFO may suffer from Belady's anomaly
- RANDOM (Any terms or words expressing the same meaning is acceptable.)
Most recently used (MRU)

File System

Explanation: The total is **5 points**, and each blank accounts for **1 point**.

- 4(4.15), 625(638)

$$\frac{60}{7200} = 8.3ms$$

. The average roationa latency is only half, so it is 4ms.

$$\frac{100MB}{160MB/s} = 625ms$$

Other acceptable solutions are provided in the brackets.

- 2 (The only correct solution.)
- Erase, program: If you answer '*clean*' or 'remove' for the first one, it is acceptable. If you answer '*write*' for the second one, it is also acceptable. All others should be graded as wrong.